Each student's Aadhaar number is unique. In this assignment, use your 12-digit Aadhaar number to generate the Arrival Time (AT) and Burst Time (BT) for 12 processes (P1–P12). Based on the generated process information, calculate the Average Turnaround Time (TAT), Average Waiting Time (WT), and Average Response Time (RT). Present your solution using a process table and a Gantt chart for each of the following CPU scheduling algorithms:

1. First Come First Serve (FCFS)
2. Shortest Job First (SJF – Non-Preemptive)
3. Shortest Remaining Time First (SRTF)
4. Priority Scheduling (Non-Preemptive)
5. Priority Scheduling (Preemptive)
6. Round Robin (Time Quantum = 3)

Instructions

1. Consider the 12 digits of your Aadhaar number in the order they appear.
2. Assign the digits sequentially as the Arrival Times (AT) of the 12 processes:
   - P1 → 1st digit
   - P2 → 2nd digit
   - …
   - P12 → 12th digit
3. Determine the Burst Times (BT) by reversing the 12 digits of your Aadhaar number. Assign the reversed digits sequentially to the corresponding processes:
   - P1 → 12th digit
   - P2 → 11th digit
   - …
   - P12 → 1st digit
4. If any digit is 0, use 0 as the corresponding Arrival Time (AT). While assigning the reversed digits as Burst Time (BT), replace every occurrence of 0 with 10.
5. For Priority Scheduling (both Preemptive and Non-Preemptive), assign priorities according to the following rule:
   - Lower Burst Time → Higher Priority.
   - If two or more processes have the same Burst Time, assign the higher priority to the process with the earlier Arrival Time. If the Arrival Times are also identical, follow the process ID order (P1, P2, …, P12).

Example

Suppose a student's Aadhaar number is:

527840613945

The generated process information is:

Process	**P1**	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**P8**	**P9**	**P10**	**P11**	**P12**
Arrival Time (AT)	5	2	7	8	4	0	6	1	3	9	4	5
Burst Time (BT)	5	4	9	3	1	6	10	4	8	7	2	5

For each scheduling algorithm, submit:

- Process table containing AT, BT, Priority (where applicable), CT, TAT, WT, and RT.
- Gantt Chart.
- Average Turnaround Time (TAT).
- Average Waiting Time (WT).
- Average Response Time (RT).
- A brief comparison of the six scheduling algorithms based on the obtained performance metrics.